How to build Data agents in Databricks
Go to Genie space & click
workspace & Default schema
Select data you create
Configure to answer your data

Under data you can add another dataset
Under instructions & you instruct it's what your dataset
used to do.

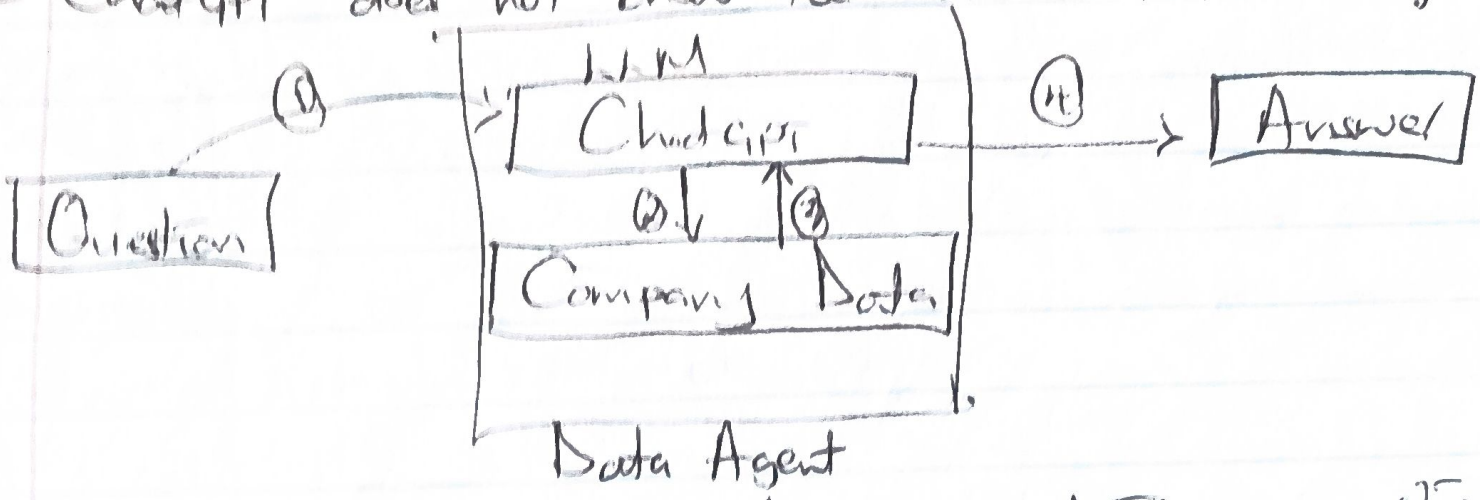
Anatomy of a Data Agent
Seven step pipeline
Question → Understand intent & write SQL → Query data
Interpret result → write answer → you follow up

Lecture

Data Agent is an AI system that can understand questions, access data, perform analysis and provide actionable answers or actions based on that data.

Question → Chat GPT → Answer

- ChatGPT does not know the data within a given Company



→ When we have a data Agent we ask it a question in natural language the question gets converted into SQL then SQL script → we query that SQL against the

Precision
Defined as the measure of the accuracy
Precision measure how many of the positive predictions made by the model were actually correct

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positive + False Positive}}$$

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{12}{12+1} = 0,91 = 91\%$$

When precision is important
Precision is important when false prediction are costly
for example
- fraud
- Spam emails
A model with high precision makes fewer incorrect positive predictions

Simple.
- Use precision when you want positive predictions to be trustworthy
- Use accuracy when you care about the overall correctness and the dataset is balanced
However, for imbalanced datasets, accuracy can be misleading, so also check precision, recall, and confusion matrix and f1-score.

Recall
is also called the "True Positive rate"

Recall $\frac{TP}{TP+FN}$

recall measures how many of the actual positive cases the model

When is recall important
Recall is important when missing a positive case is costly

- Detecting diseases
- fraud
- Customer churn

} A model with high recall produces fewer false negatives

F1-score
measure the models accuracy
Combines the precision and recall scores of a model

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

$$F1 = \frac{2TP}{2TP + fp + fN}$$

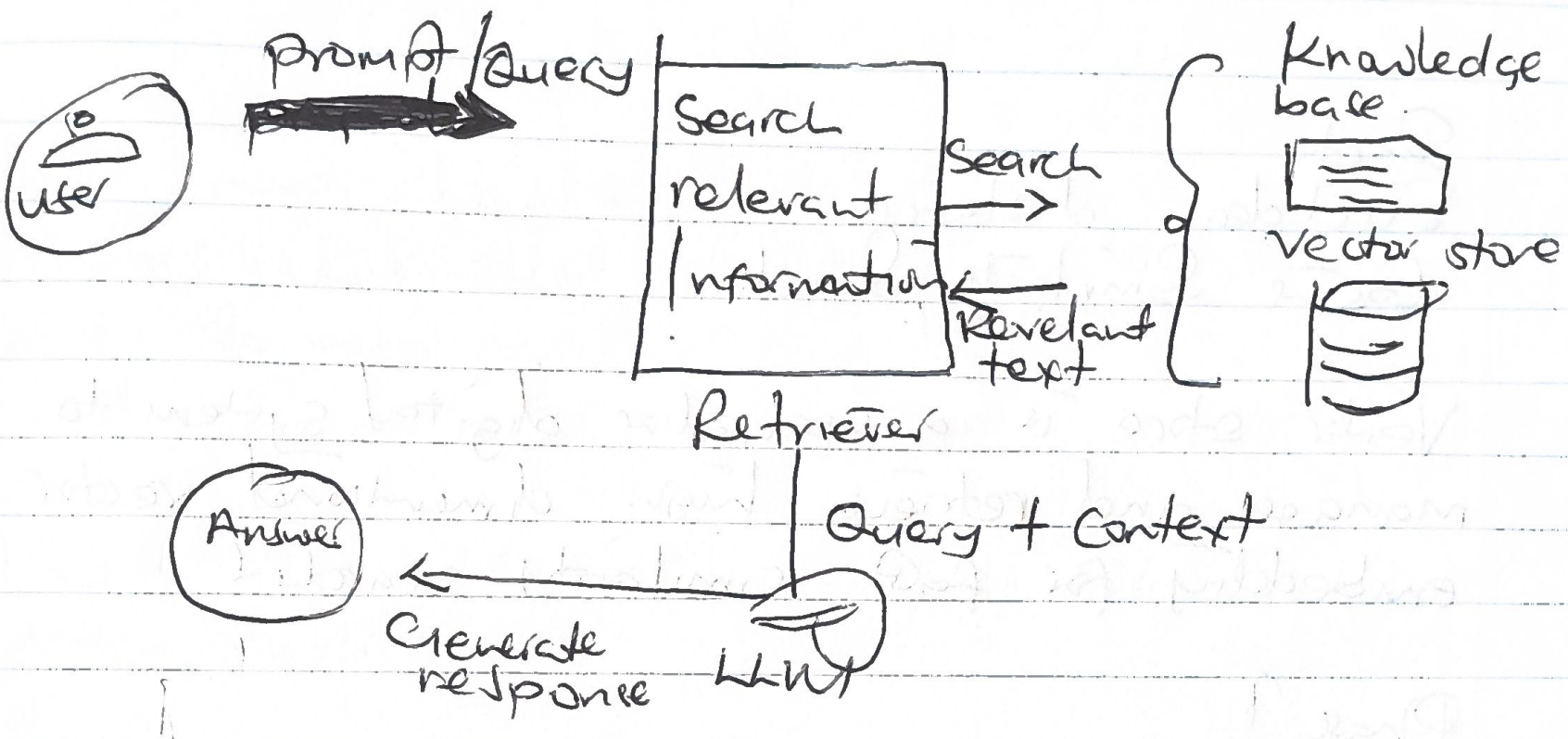
Calculating it using the confusion matrix
When is the F1-score important
The dataset is imbalanced
Both false positive (FP) and false negative (FN)
- H balanced precision and recall

Interpretation: F1 = 1 ; Perfect precision and recall

f1 close to 0: Poor model perform
A high f1-score means both precision and recall are high

RAG (Retrival Augmented Generation

- It is a method where an AI system first retrieved relevant information and then generate an answer using that information



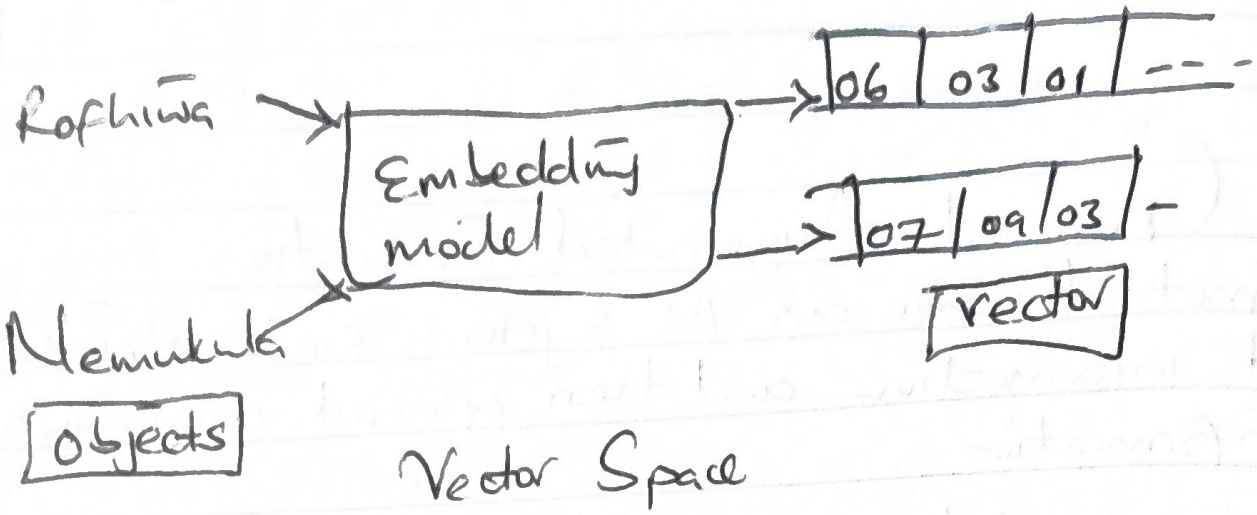
Without RAG - Answering generally with RAG - checking notes, then answering.

Vectors.
How RAG works

Vector Embedding

Is a numerical representation of data points that represents different data types - including non-mathematical data such as words or images

- It can be an array of numbers that a machine learning model can process



Vector Space

Faetr
Euclidean distance
Cosine Similarity Search

Vector store is a specialize digital system to store manage and retrieve high dimensional vector embedding for fast similarity searches

Phase 1

Data Sources
Ingestion
Extraction
Cleaning
Chunking
Embedding
Vector DB

